Sample outputs

Create team

```
========== MAIN MENU ==========
1. Create New Team
2. Manage Team (Select Team)
3. List All Teams
4. Simulate Match (Team vs Team)
5. Exit
================================
Enter choice: 1

--- CREATE NEW TEAM ---
Enter team name: orlando
Enter team manager name: mike shabangu
SUCCESS: Team 'orlando' has been created!
Manager: mike shabangu
```

Manage team

```
Enter choice: 2

--- SELECT TEAM ---
1. FC Barcelona (Manager: Xavi Hernandez) | Players: 11/11
2. Manchester City (Manager: Pep Guardiola) | Players: 11/11
3. Real Madrid (Manager: Carlo Ancelotti) | Players: 11/11
4. orlando (Manager: mike shabangu) | Players: 0/11
Enter choice (0 to cancel): 4
SUCCESS: Now managing team 'orlando'
Manager: mike shabangu

========== TEAM MANAGEMENT ==========
Current Team: orlando
Manager: mike shabangu
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: |
```

Add player

```
========== TEAM MANAGEMENT ==========
Current Team: orlando
Manager: mike shabangu
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
======================================
Enter choice: 1

--- ADD NEW PLAYER TO orlando ---
Team Manager: mike shabangu
Enter player name: prince
Enter jersey number (1-99): 45
Enter skill rating (0-100): 78

Select position:
1. Goalkeeper
2. Defender
3. Midfielder
4. Striker
Choice: 4
SUCCESS: prince (#45) added to orlando!
Team Manager: mike shabangu approved the signing.
```

Display Team Roster

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
======================================
Enter choice: 2

========================================
TEAM: Real Madrid | MANAGER: Carlo Ancelotti
========================================[GK]  Thibaut Courtois | #1  | Skill: 93
[DF]  Dani Carvajal   | #2  | Skill: 85
[DF]  Antonio Rudiger | #22 | Skill: 88
[DF]  David Alaba     | #4  | Skill: 87
[DF]  Ferland Mendy   | #23 | Skill: 84
[MF]  Luka Modric     | #10 | Skill: 90
[MF]  Toni Kroos      | #8  | Skill: 89
[MF]  Eduardo Camavinga | #12 | Skill: 85
[MF]  Federico Valverde | #15 | Skill: 88
[ST]  Vinicius Junior | #7  | Skill: 90
[ST]  Jude Bellingham | #5  | Skill: 91

Average Skill: 88.2
========================================
```

Apply Formation Strategy

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
======================================
Enter choice: 3

--- SELECT FORMATION FOR Real Madrid ---
Team Manager: Carlo Ancelotti is making tactical decision.
1. 4-4-2 (Balanced)
2. 4-3-3 (Attacking)
3. 3-5-2 (Possession)
Choice: 1
Applying 4-4-2: Balanced tactic.
SUCCESS: Formation applied to Real Madrid!
Manager Carlo Ancelotti has set the team's formation.

Recommended formation based on squad: 4-4-2
```

Sort Team Players

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
======================================
Enter choice: 4

Sort players by:
1. Skill (highest first) - Bubble Sort O(n^2)
2. Jersey Number - Quick Sort O(n log n)
Choice: 1
Team sorted by skill rating using BUBBLE SORT!
Manager Carlo Ancelotti requested skill-based sorting.

========================================
TEAM: Real Madrid | MANAGER: Carlo Ancelotti
========================================[GK]  Thibaut Courtois | #1  | Skill: 93
[ST]  Jude Bellingham | #5  | Skill: 91
[MF]  Luka Modric     | #10 | Skill: 90
[ST]  Vinicius Junior | #7  | Skill: 90
[MF]  Toni Kroos      | #8  | Skill: 89
[DF]  Antonio Rudiger | #22 | Skill: 88
[MF]  Federico Valverde | #15 | Skill: 88
[DF]  David Alaba     | #4  | Skill: 87
[DF]  Dani Carvajal   | #2  | Skill: 85
[MF]  Eduardo Camavinga | #12 | Skill: 85
[DF]  Ferland Mendy   | #23 | Skill: 84
```

```
========================================
Enter choice: 4

Sort players by:
1. Skill (highest first) - Bubble Sort O(n^2)
2. Jersey Number - Quick Sort O(n log n)
Choice: 2
Team sorted by jersey number using QUICK SORT!
Manager Carlo Ancelotti requested number-based sorting.

========================================
TEAM: Real Madrid | MANAGER: Carlo Ancelotti
========================================[GK]  Thibaut Courtois | #1  | Skill: 93
[DF]  Dani Carvajal   | #2  | Skill: 85
[DF]  David Alaba     | #4  | Skill: 87
[ST]  Jude Bellingham | #5  | Skill: 91
[ST]  Vinicius Junior | #7  | Skill: 90
[MF]  Toni Kroos      | #8  | Skill: 89
[MF]  Luka Modric     | #10 | Skill: 90
[MF]  Eduardo Camavinga | #12 | Skill: 85
[MF]  Federico Valverde | #15 | Skill: 88
[DF]  Antonio Rudiger | #22 | Skill: 88
[DF]  Ferland Mendy   | #23 | Skill: 84

Average Skill: 88.2
========================================
```

5. Search for Player

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
========================================
Enter choice: 5

Search for player:
1. By Name (Linear Search - O(n))
2. By Jersey Number (Binary Search - O(log n))
Choice: 1
Enter name to search: ToniH
Player not found!
```

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 5

Search for player:
1. By Name (Linear Search - O(n))
2. By Jersey Number (Binary Search - O(log n))
Choice: 1
Enter name to search: Toni Kroos
Player found using LINEAR SEARCH: [MF]  Toni Kroos      | #8  | Skill: 89
```

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 5

Search for player:
1. By Name (Linear Search - O(n))
2. By Jersey Number (Binary Search - O(log n))
Choice: 2
Enter jersey number: 10
Player found using BINARY SEARCH: [MF]  Luka Modric     | #10 | Skill: 90
```

```
========== TEAM MANAGEMENT ==========
Current Team: Real Madrid
Manager: Carlo Ancelotti
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 5

Search for player:
1. By Name (Linear Search - O(n))
2. By Jersey Number (Binary Search - O(log n))
Choice: ytv
ERROR: Invalid choice.
```

Simulate Match (Team vs Team)

```
========== MAIN MENU ==========
1. Create New Team
2. Manage Team (Select Team)
3. List All Teams
4. Simulate Match (Team vs Team)
5. Exit
===============================
Enter choice: 4

--- SIMULATE MATCH ---
1. FC Barcelona (Manager: Xavi Hernandez) | Players: 11/11 | READY
2. Manchester City (Manager: Pep Guardiola) | Players: 11/11 | READY
3. Real Madrid (Manager: Carlo Ancelotti) | Players: 11/11 | READY
4. orlando (Manager: mike shabangu) | Players: 1/11 | NOT READY

Select HOME team (0 to cancel): 1
Select AWAY team (0 to cancel): 2

========================================
MATCH: FC Barcelona vs Manchester City
Home Manager: Xavi Hernandez
Away Manager: Pep Guardiola
========================================

Final Score: FC Barcelona 8 - 7 Manchester City
Manager Xavi Hernandez: 'Great performance from the team!'
Manager Pep Guardiola: 'We need to improve for next match.'
```

```
--- SIMULATE MATCH ---
1. FC Barcelona (Manager: Xavi Hernandez) | Players: 11/11 | READY
2. Manchester City (Manager: Pep Guardiola) | Players: 11/11 | READY
3. Real Madrid (Manager: Carlo Ancelotti) | Players: 11/11 | READY
4. orlando (Manager: mike shabangu) | Players: 1/11 | NOT READY

Select HOME team (0 to cancel): 3
Select AWAY team (0 to cancel): g

========== MAIN MENU ==========
1. Create New Team
2. Manage Team (Select Team)
3. List All Teams
4. Simulate Match (Team vs Team)
5. Exit
===============================
Enter choice: ERROR: Invalid input. Please enter a number (1-5).
```

List all teams

```
========== MAIN MENU ==========
1. Create New Team
2. Manage Team (Select Team)
3. List All Teams
4. Simulate Match (Team vs Team)
5. Exit
================================
Enter choice: 3

========== ALL TEAMS ==========
1. Team: FC Barcelona
   Manager: Xavi Hernandez
   Players: 11/11
   Total Skill: 954
   Average Skill: 86.7
   Formation: 4-3-3
   Can Play Match: Yes
   Full Squad: Yes
----------------------------------------
2. Team: Manchester City
   Manager: Pep Guardiola
   Players: 11/11
   Total Skill: 975
   Average Skill: 88.6
   Formation: 3-5-2
   Can Play Match: Yes
   Full Squad: Yes
----------------------------------------
3. Team: Real Madrid
   Manager: Carlo Ancelotti
   Players: 11/11
   Total Skill: 970
   Average Skill: 88.2
```

ERROR HANDLING

```
========== MAIN MENU ==========
1. Create New Team
2. Manage Team (Select Team)
3. List All Teams
4. Simulate Match (Team vs Team)
5. Exit
================================
Enter choice: 55
ERROR: Invalid choice. Please select 1-5.
```

```
Enter choice (0 to cancel): 1
SUCCESS: Now managing team 'FC Barcelona'
Manager: Xavi Hernandez

========== TEAM MANAGEMENT ==========
Current Team: FC Barcelona
Manager: Xavi Hernandez
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: Y
ERROR: Invalid input.
```

```
========== TEAM MANAGEMENT ==========
Current Team: FC Barcelona
Manager: Xavi Hernandez
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 4

Sort players by:
1. Skill (highest first) - Bubble Sort O(n^2)
2. Jersey Number - Quick Sort O(n log n)
Choice: 3
ERROR: Invalid choice.
```

```
Current Team: FC Barcelona
Manager: Xavi Hernandez
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 3

--- SELECT FORMATION FOR FC Barcelona ---
Team Manager: Xavi Hernandez is making tactical decision.
1. 4-4-2 (Balanced)
2. 4-3-3 (Attacking)
3. 3-5-2 (Possession)
Choice: 7
ERROR: Invalid formation choice.
```

```
========== TEAM MANAGEMENT ==========
Current Team: FC Barcelona
Manager: Xavi Hernandez
1. Add Player to Team
2. Display Team Roster
3. Apply Formation Strategy
4. Sort Team Players
5. Search for Player
6. Back to Main Menu
=====================================
Enter choice: 4

Sort players by:
1. Skill (highest first) - Bubble Sort O(n^2)
2. Jersey Number - Quick Sort O(n log n)
Choice: U
ERROR: Invalid choice.
```